

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

**Duration : 45 Minutes
Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

I B. Theerthi (192473026) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

B. Theerthi
Signature of the student:

16
—
20

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

2. A robot is placed in an unknown maze using UCS and IDS

Problem statement

A robot is placed in an unknown maze and must reach the exit without knowing the layout.

The robot should explore the maze and find a path

Uniform Cost Search

Uniform cost search expands the node with the lowest cumulative path cost

Ex:-

$A \rightarrow B \rightarrow \text{Goal} = \text{cost } 10$

$C \rightarrow D \rightarrow \text{Goal} = \text{cost } 16$

UCS select the path 1 because cost is low.

Advantages

- complete
- Finds the least - cost path
- Suitable when movement cost is vary

Disadvantages

High memory usage
Slow for large mazes

Time : $O(b^{(1+d/\epsilon)})^*$

Space : $O(b^{(1+c/\epsilon)})^*$

Where

b = branching factor

c^* = optimal path cost

ϵ = minimum step cost

Iterative Deeping Search

IDS repeatedly performs Depth-limited search with increasing depth limit

Ex: Depth 0

Depth 1

Depth 2

continue until goal reached

Advantages

- Low memory usage
- complete
- Suitable when depth is unknown

DisAdvantages

- Repeat node expansion
- Not optimal if step cost differ

Complexity

Time : $O(b^d)$

Space : $O(bd)$

Where

$b \rightarrow$ branching factor

$d \rightarrow$ depth of solution

1. A smart home automation system controls lighting, temp and security using sensor data and user preferences. Different intelligent can be used:

• Simple Reflex agent

- Works only on the current input using Condition action rules

Example

- If model is detected $\rightarrow$ Turn on lights
- If temp above 30°C $\rightarrow$ Turn on Ac

Advantage :- Fast and simple

Disadvantage :- Does not remember past events or learn user habits

2. Model-based Reflex Agent

- Maintain a internal model of the environment
- use current sensor data and previous information

Example

- Known a room was occupied recently so it keeps lights on even if motion stops briefly

Advantage: Better decisions in partially observable environment

Disadvantage:- Most complex than a simple reflex agent

Goal-Based Agent

- choose action that ~~will~~ achieve a specific goal
- Example:
Goals: maintain room temp at 24°C while saving energy
- Advantage: Flexible
- DisAdvantage: Require more computation

Utility-Based Agent

- selects the action with highest overall benefit
- Balances comfort, energy efficiency
- Example
 - Adjust AC and lighting to maximize comfort while minimizing electricity usage
- Advantage: Makes optimal decisions with multiple objectives
- DisAdvantage: utility calculation can be complex

Hybrid approach will be the more efficient for ensuring comfort energy efficiency and security